



# ZEPH

Decentralised flight delay compensation, enforced on-chain.

01 / THE PROBLEM

EU Regulation 261/2004 entitles passengers to compensation for delays of 3+ hours. In practice, airlines make these claims opaque, slow, and easy to ignore. Most eligible passengers never receive what they're owed.

02 / HOW IT WORKS

01

Register

Passenger registers their flight on-chain via the Zeph smart contract.

02

Monitor

A custom oracle polls flight data every 60 seconds and reports delays  $\geq$  180 minutes.

03

Decide

The airline accepts or rejects the claim on-chain, attaching an evidence hash.

04

Auto-accept

If the airline doesn't respond within 7 days, the claim is automatically accepted.

03 / TECH STACK

|    |          |                 |    |           |               |
|----|----------|-----------------|----|-----------|---------------|
| 01 | Solidity | SMART CONTRACT  | 02 | Hardhat   | CONTRACT DEV  |
| 03 | Polygon  | L2 CHAIN        | 04 | Ethers.js | WEB3 CLIENT   |
| 05 | Next.js  | FRONTEND        | 06 | React     | UI RUNTIME    |
| 07 | Supabase | OFF-CHAIN STORE | 08 | MetaMask  | CRYPTO WALLET |

04 / KEY FEATURES

- ◆

On-chain evidence hashing for a tamper-proof audit trail.
- ◆

PDF / ZIP evidence export for regulatory escalation.
- ◆

Role-based portals: Passenger · Airline · Admin.
- ◆

Reentrancy-protected smart contract with role-based access control.